Matthew McKay, Ph.D.

# Agency nurses vs. staff nurses

**Case study C · A workforce cost model: where converting agency hours to staff positions would pay, with sensitivity ranges**

*Tessaly Health is a fictional nonprofit health system. Staffing data is synthetic. Cost inputs are national public benchmarks or labeled assumptions. Results show a method, not real findings.*

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## 1. The question

Tessaly Health spent about $6.8 million on agency nurses in 2025. Leadership asked the obvious question: would hiring staff nurses for those hours be cheaper, and where? The answer needed to say how sure it was, because a conversion program means hiring people.

## 2. The data

| Source | What | Size |
|---|---|---|
| Synthetic generator (LIB-23, seed 20261108) | Required, staff, overtime, agency and unfilled RN hours by inpatient unit and day | 18 units × 1,096 days |
| Same | Agency contracts; RN hires, exits and time to proficiency | 634 contract segments; 6,387 employment records |
| BLS Employer Costs for Employee Compensation, Dec 2024 | Staff RN wage per hour worked ($51.09) and benefits ($28.43) in private hospitals | National |
| SIA/NATHO Travel Nurse Benchmarking Survey | Agency bill rates: $106.63 (2023), $90.13 (2024), $90.54 (2025) | National |
| NSI National Health Care Retention & RN Staffing Report 2026 | Time to recruit an experienced RN (78 days, range 56–102) | National |
| My assumptions | Cost per hire ($12,000), productivity while ramping (50%), worked hours per FTE (1,700), overtime payroll load (7.65%) | Each varied in the sensitivity analysis |

Turnover, exit rates and time to proficiency come from the synthetic data, which is tuned to national surveys.

**Limits.** The model prices cost only. Continuity of care, team knowledge and overtime fatigue are real and left out; section 6 says how I'd handle them. Benchmarks are national averages, and a real system would use its own rates.

## 3. Approach

1. **Compare like with like.** Cost per *worked* hour for staff (wage plus benefits), overtime, and agency (bill rate plus orientation fee). Agency nurses also ramp up when they start on a unit. With assignments averaging 116 days and 13 days to proficiency, that adds about 6% to the effective agency rate. Leaving it out would tilt the comparison toward conversion.
2. **Find the base, not the average.** A staff nurse is paid every scheduled shift, whether the hours are needed or not. So I split each unit's daily agency hours into slices of one full-time position and measured from the 2025 data how often each slice was actually needed. The first slice on a busy unit is needed every day; the fifth may be needed only 80% of the time.

3. **Simulate each position for 3 years.** Run day by day: recruiting lag, 14 days of orientation, a ramp to proficiency, exits at the unit's observed rates, refills, and agency covering every gap. That's 400 runs per position, with uncertain inputs drawn from ranges.
4. **Classify each position.** *Convert* if at least 80% of runs save money; *pilot* if 60–80% save money and the average is positive; otherwise *keep on agency*.
5. **Test sensitivity.** Move each input across its range for one representative position.

**Alternatives I considered.** A simple hourly comparison ($93 vs. $83, "save $10 an hour") is what usually gets presented, and it overstates the case by ignoring hiring, ramp, turnover and idle hours. A full staffing-optimization model would be more precise, but it needs inputs (patient volume, skill mix) that a cost question shouldn't wait for.

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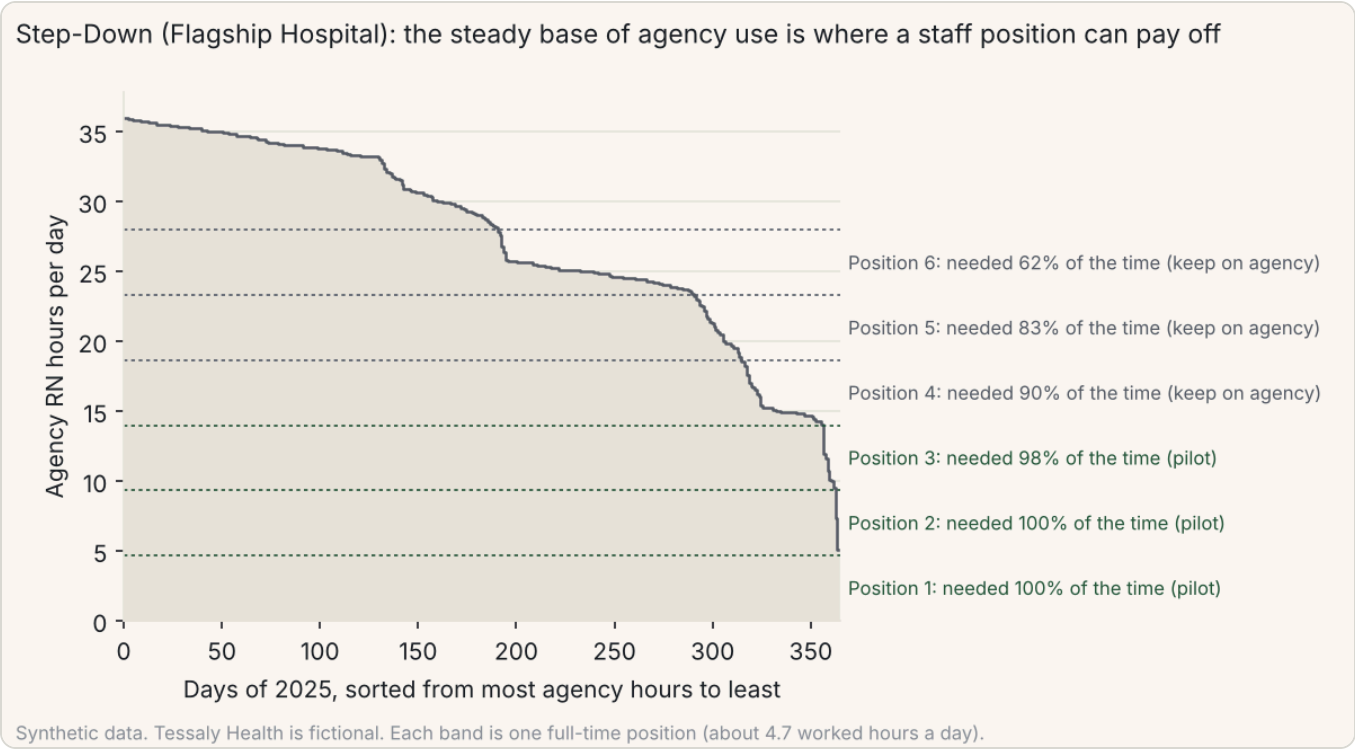
## 4. Validation

| Check | Result |
|---|---|
| With no exits and full use, the simulation matches a hand calculation | $37,879 vs. $37,903. Pass |
| Loaded staff rate read back against the BLS benchmark it came from | $79.70 vs. $79.53 for 2024 (rounding the benefits load). Pass |
| Converted positions never cover more hours than the unit bought from agency | 41 positions on 13 units. Pass |
| Results stable with a different random seed | 39 of 41 classifications identical; the 2 that changed sit at the 60% line. Pass |
| The LIB-26 calculator (JavaScript) reproduces the Python model | $13,979 vs. $13,322 for the same inputs (within simulation noise). Pass |

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## 5. Findings (synthetic data, national benchmark rates)

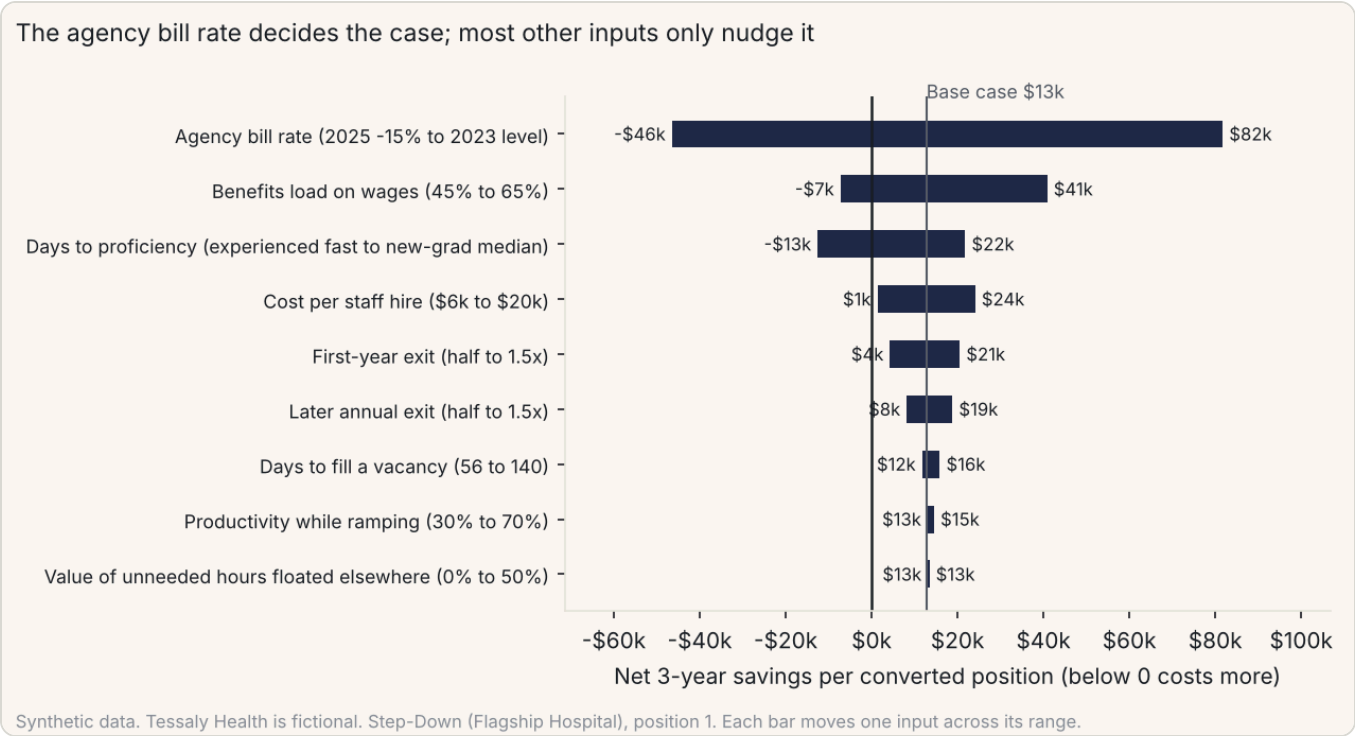
**The hourly gap is smaller than people think.** In 2025 a fully loaded staff hour cost $83.05 and an agency hour $93.10: a 12% premium, or about 19% once agency ramp-up is counted. In 2023, at $106.63 an hour, the premium was 43%. The case for conversion was strong two years ago and is thin now.




**Only the steady base can pay.** Of 41 possible positions on 13 units, none clears the “convert” bar and 8 qualify for a pilot, on five high-turnover units. Each pilot position has an expected 3-year saving of $7,000 to $16,000 and a 60–67% chance of saving anything. Any position needed less than about 90% of the time loses money, because the hospital pays for hours it doesn’t use.

| Unit | Pilot positions | Expected 3-year saving |
|---|---|---|
| Medical-Surgical, Community North | 2 | $29,000 |
| Step-Down, Flagship Hospital | 3 | $28,500 |
| Medical-Surgical 1, Flagship Hospital | 1 | $16,200 |
| Medical-Surgical, Community South | 1 | $13,000 |
| Emergency Department, Community South | 1 | $9,100 |
| **Total** | **8** | **about $96,000 (about $32,000 a year)** |

That’s under 0.5% of annual agency spend. It’s real, but it isn’t a program.




**The agency rate decides it.** The break-even agency bill rate is about **$87.40 an hour**, just $3 below the 2025 benchmark. At the 2023 rate, the same position saves $84,000 over 3 years. Benefits cost and time to proficiency come next. A new graduate's 168-day ramp, against 74 days for an experienced hire (case study B), turns a small saving into a loss.

**Rate negotiation is worth more than conversion.** A $1-an-hour reduction on Tessaly's 73,500 agency hours saves about $73,500 a year, more than twice what all eight pilot positions are expected to save.

**Retention helps, a little.** If on-time onboarding cut first-year exits as case study B suggests (risk ratio 0.77), the first Step-Down position's expected saving rises from $12,800 to $15,700. Worth having, not decisive.

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## 6. What I'd do in practice

- **Don't announce a conversion program.** Pilot the 3–4 steadiest positions (Step-Down at Flagship and Medical-Surgical at Community North), filled with **experienced** nurses, and track them against similar units for a year.
- **Negotiate the rate first.** It's the largest lever in the model and needs no hiring.
- **Set a trigger to revisit.** Rerun the model when the agency bill rate passes about $95 an hour, or when benefits costs or turnover change materially. The calculator makes that a five-minute check.
- **Decide the non-cost factors in the open.** If nursing leadership values continuity enough to convert positions that are cost-neutral, that's a legitimate choice. It should be recorded as a choice, not hidden in an assumption.
- **Work on retention anyway.** Every point of first-year turnover avoided makes every staff position cheaper, whether or not it replaced agency.

The calculator ( `calculator.html` , LIB-26) lets anyone change the inputs and see the result.

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## 7. Files

| File | What it is |
|---|---|
| `calculator.html` | LIB-26 conversion calculator (single web page, no external code) |
| Agency-to-staff conversion calculator (LIB-26) | LIB-26 page: purpose, worked example, design notes |
| `analysis/convert.py` | Rates, spend, slices, simulation, classification, tornado, scenarios |
| `analysis/visuals_c.py` | Charts |
| `analysis/validate_c.py` | Hand-calculation check, benchmark read-back, coverage check, seed stability, calculator cross-check |
| `outputs/` | Result tables, inputs taken from the data, figures, validation report |

To reproduce:

```
cd case-studies/c-agency-vs-staff/analysis
python convert.py && python visuals_c.py && python validate_c.py
```

*All code is AI-assisted (Claude). I designed the cost comparison, the base-load method, the simulation and the checks; Claude wrote the code to that design; the validation above is how I checked it.*

**Sources:** BLS, Charting compensation costs during National Nurses Month (2025); SIA/NATHO Travel Nurse Benchmarking Survey, selected findings 2025 and 2026; NSI Nursing Solutions, *2026 National Health Care Retention & RN Staffing Report*, as summarized by Becker's.

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Personal project. All data is synthetic or public, and the company is fictional. Views are my own and not affiliated with or endorsed by any current or former employer.